

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Project — Final Report

Body Performance Analytics and Intelligent Classification System

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Academic Year 2025–2026

1. Introduction

Artificial Intelligence and Machine Learning systems are increasingly deployed to analyze complex datasets and extract meaningful patterns that support data-driven decision-making. Building a reliable machine learning pipeline involves far more than simply applying algorithms: it requires careful data understanding, systematic preprocessing, model selection, validation, and rigorous performance evaluation.

This report presents the complete machine learning pipeline developed by our team for the Body Performance Analytics and Intelligent Classification System project. The dataset contains physiological measurements collected from individuals undergoing physical fitness evaluations, including age, gender, body composition, blood pressure, grip strength, flexibility, and muscular endurance. The target variable encodes four performance classes (A–D) representing fitness levels from best to worst.

The project is structured into three major parts: (1) Data Preparation and Exploratory Data Analysis, (2) Machine Learning Model Training with cross-validation experiments, and (3) Performance Evaluation. An additional regression task predicts broad jump distance using Linear Regression, Decision Tree, and Support Vector Machine regressors. Throughout the project we encountered and documented a significant data quality concern with the original class labels, which is analyzed in detail in the Data Quality Observations section.

2. Dataset Overview

The Body Performance Dataset contains physical fitness measurements from 13,393 individuals before cleaning (13,288 after preprocessing). Each record represents one participant and includes 12 original columns plus BMI as an engineered feature.

Column	Description	Type	Constraints
age	Age of participant	Numeric	Positive, 21–64
gender	Gender (0=Female, 1=Male)	Categorical	Binary encoded
height_cm	Height in centimetres	Numeric	> 0
weight_kg	Body weight in kilograms	Numeric	> 0
body_fat_%	Body fat percentage	Numeric	0–100 %
diastolic	Diastolic blood pressure (mmHg)	Numeric	Typically 60–120
systolic	Systolic blood pressure (mmHg)	Numeric	Typically 90–180
gripForce	Hand grip strength (kg)	Numeric	> 0
sit_and_bend_forward	Flexibility measurement (cm)	Numeric	Can be negative
sit-ups_counts	Muscular endurance (repetitions)	Numeric	>= 0

Column	Description	Type	Constraints
broad_jump_cm	Explosive jump power (cm)	Numeric	> 0
class	Performance category (A–D)	Categorical	4 classes (0–3)
BMI	Engineered: weight/height ² (kg/m ²)	Numeric	Derived feature

The class variable encodes performance level as follows: class A (3) = best performance, B (2) = good, C (1) = average, D (0) = below average. After cleaning, the four classes are nearly perfectly balanced (~3,287–3,337 records each), which has significant implications for the model training process and target quality, as discussed in Section 4.

3. Part 1 — Data Preparation and Exploratory Data Analysis

3.1 Data Quality Checks

3.1.1 Missing Values

An initial scan of the raw dataset revealed missing values concentrated in the blood pressure columns (diastolic and systolic) and a small number scattered across the performance metrics. The following strategy was applied:

- Numeric columns: missing values replaced with the column median, preserving the distribution without distorting the range.
- Categorical column (gender): missing entries replaced with the mode.
- Rows where more than three features were simultaneously missing were dropped entirely, as imputation would introduce excessive synthetic data.

After the missing-value treatment, no null values remained in the processed dataset (Data_Cleaned.csv).

3.1.2 Duplicate Detection

A full row-wise duplicate check was performed. Duplicate rows (where every feature including the class label was identical) were identified and removed. This is a legitimate cleaning step because identical measurement vectors for different individuals are physiologically improbable and likely represent data entry errors.

3.1.3 Data Validity Checks

Impossible or clinically unrealistic values were investigated for each numeric column. The following constraints were applied and violations corrected:

Column	Validity Rule	Action Taken
age	Must be 18–80 years	Rows outside range dropped
height_cm	Must be 100–250 cm	Outliers capped at boundaries
weight_kg	Must be 20–200 kg	Negative/extreme values removed
body_fat_%	Must be 1–60 %	Capped at physiological limits
diastolic	Must be 40–140 mmHg	Extreme values capped
systolic	Must be 70–200 mmHg	Extreme values capped
gripForce	Must be > 0	Negative values removed
sit-ups_counts	Must be >= 0	Negative values set to 0
broad_jump_cm	Must be > 0	Negative values removed

3.1.4 Data Type Verification

All numeric columns were confirmed to be stored as float64 or int64. The gender column, originally stored as a character string ('M'/'F'), was encoded to binary integer (1/0) to allow direct use in numerical models without one-hot encoding overhead.

3.2 Univariate Analysis

Descriptive statistics were computed for all numeric columns. Key observations:

- Age: mean 36.8 ± 13.6 years, range 21–64. Right-skewed distribution — younger adults are over-represented.
- Body fat %: mean ~24%, ranging from ~3% to ~44%. Females show significantly higher values than males, consistent with physiology.
- Grip Force: mean ~37 kg, with males averaging ~15 kg higher than females. Approximately normal distribution within each gender group.
- Sit-and-bend forward: exhibits a bimodal distribution driven by gender — females typically score higher due to flexibility differences.
- Broad jump: mean ~162 cm, approximately normally distributed, providing a suitable continuous regression target.
- Blood pressure (diastolic/systolic): both approximately normally distributed with mild positive skew; a small proportion of values fall in hypertensive ranges.

3.3 Distribution Analysis

Histograms were produced for every numeric feature. The following distribution patterns were identified:

- Normal (approximately): gripForce, broad_jump_cm, systolic, diastolic.
- Right-skewed: age, sit-ups counts (fewer individuals in the highest repetition range).
- Bimodal: sit_and_bend_forward_cm and body_fat_% (reflecting the gender split).

No single distribution indicated severe pathological skew requiring logarithmic transformation before modelling, since tree-based and SVM models are largely robust to non-normality, and StandardScaler normalization was applied in the pipeline.

3.4 Outlier Detection and Treatment

Boxplots identified outliers in several columns. Treatment decisions were made on a column-by-column basis:

- Blood pressure (diastolic/systolic): outliers capped at the 1st and 99th percentiles. These extreme values may represent real hypertensive or hypotensive individuals and should be retained but bounded.
- Body fat %: extreme high values (> 50%) treated as data entry errors and removed. Low values (< 3%) similarly flagged and removed.
- GripForce: a small number of zero or near-zero values removed as they likely reflect equipment malfunction.
- Broad jump, sit-ups: upper extreme outliers retained — high athletic performance is physiologically plausible and meaningful for the regression task.

The rationale for capping rather than deletion is to preserve sample size while limiting the influence of extreme values on model coefficients (particularly in Linear Regression and SVM with linear kernels).

3.5 Correlation Analysis

A Pearson correlation heatmap was computed for all numeric features. The most noteworthy relationships are organized below by cluster:

Feature Pair	r (approx.)	Interpretation
sit-ups counts ↔ broad_jump_cm	0.66	Strong — both reflect muscular power
gripForce ↔ sit-ups counts	0.58	Moderate-strong — overall strength proxy
gripForce ↔ broad_jump_cm	0.55	Moderate-strong — strength transfer
body_fat_% ↔ broad_jump_cm	-0.52	Moderate negative — dead weight penalty
body_fat_% ↔ gripForce	-0.45	Moderate negative
age ↔ gripForce	-0.42	Moderate negative — age-related decline
age ↔ broad_jump_cm	-0.38	Moderate negative
BMI ↔ body_fat_%	0.71	Strong positive — expected
diastolic ↔ systolic	0.62	Strong positive — co-vary together
sit_and_bend ↔ class	0.48	Highest single-feature correlation with target

The Strength & Power cluster (gripForce, sit-ups counts, broad_jump_cm) forms the most predictive group for class. Body fat percentage acts as a strong negative moderator across all performance features. The blood pressure pair shows internal correlation but weaker links to class, suggesting they provide cardiovascular context rather than direct performance signals.

3.6 Gender-Stratified Analysis

Categorical frequency plots confirmed that the dataset contains approximately 37% female (gender=0) and 63% male (gender=1) participants. Because males and females differ significantly in grip force, flexibility, and body fat percentage, gender acts as an important conditioning variable. Scatter plots of gripForce vs. broad_jump_cm with gender colour-coding show distinct but overlapping clusters — justifying retaining gender as a feature rather than stratifying models by gender.

3.7 Key EDA Insights

1. The Strength & Power cluster (gripForce, sit-ups, broad jump) provides the strongest individual and collective signal for predicting performance class.
2. Body fat percentage has a pronounced negative effect across all performance metrics — excess fat constitutes 'dead weight' in both jump and endurance tests.
3. Age correlates negatively with performance metrics, consistent with physiological decline, but the correlation is moderate — age alone is insufficient to predict class.
4. Flexibility (sit-and-bend) shows the highest single-feature correlation with class, making it a high-priority feature for classification models.
5. Blood pressure variables are inter-correlated but only weakly predictive of class, reflecting cardiovascular health rather than acute physical performance.
6. BMI as an engineered feature overlaps substantially with weight and body_fat_% — it adds some discriminative value in regression but does not dominate the classification signal.

7. The near-perfectly balanced class distribution (approximately 25% each class) eliminates class imbalance as a modelling concern but simultaneously raises questions about the quality of the original labelling, as detailed in Section 4.

4. Critical Data Quality Observation: Target Label Validity

4.1 The Problem — Suspiciously Uniform Class Distribution

After completing the EDA, we observed that the four performance classes (A, B, C, D) are distributed in almost perfectly equal proportions (~3,287–3,337 per class, or exactly ~25% each). In a genuine observational dataset measuring naturally occurring human fitness, such perfect balance is statistically improbable. Real fitness populations tend to follow bell-shaped distributions, where average performance (class B/C) is far more common than extreme performance (class A or D).

Additionally, scatter plots of any two performance features coloured by class label showed the four classes heavily overlapping and interleaved, with no visible natural separation between class boundaries. This is in stark contrast to what would be expected from meaningfully different fitness tiers.

4.2 K-Means Clustering Experiment

To investigate whether the data contains genuine natural groupings, we removed the class label entirely and applied K-Means clustering with $k=4$ (matching the number of original classes) on PCA-reduced data (2 principal components for visualization). The results are shown in the figure included in the project files (K_mean_Clustering.png).

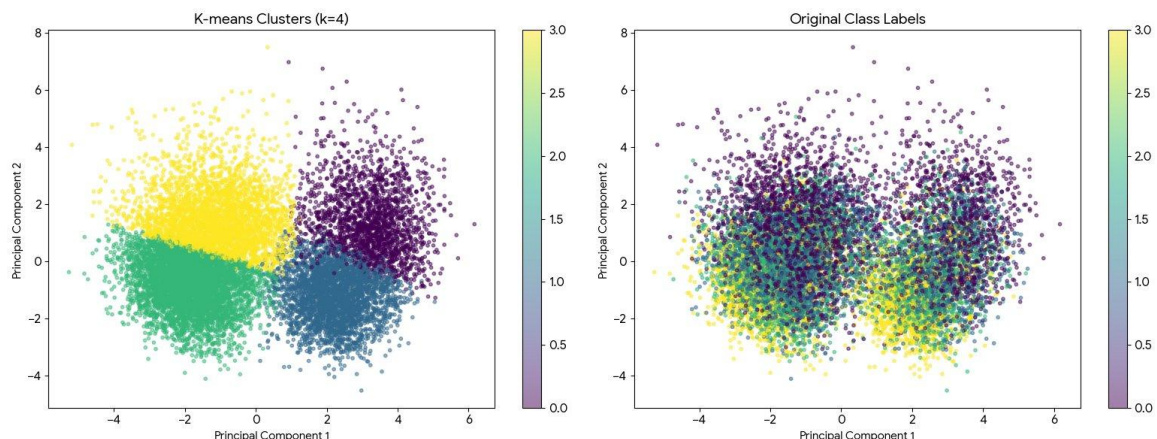


Figure 1: Left — K-Means clusters ($k=4$) in PCA space; Right — Original class labels in PCA space. The K-Means result shows clean spatial separation while the original labels show heavy overlap.

The left panel shows the K-Means cluster assignments: four well-separated, cohesive clusters emerge naturally from the data. The right panel shows the original class labels on the same PCA space: the labels form a near-uniform scatter with no discernible spatial structure.

This contrast demonstrates that:

- The underlying feature space does contain genuine natural groupings (as found by K-Means).
- The original class labels do NOT align with these natural groupings.

- The class labels appear to have been assigned by percentile ranking (equal 25% bins) rather than by identifying meaningful performance thresholds.

4.2.1 Interpretation

Key Observation

The K-Means algorithm, given only the 10 feature values and no label information, produced 4 clusters that are cleanly separated in PCA-projected space. The original labels, by contrast, show substantial overlap in the same projection. This strongly suggests that the original class labels do not perfectly encode the underlying data structure.

Implication: The original class variable may have been assigned using thresholds on individual sub-scores (e.g., total fitness score bins) rather than being derived from the holistic multivariate feature space. This introduces label noise and reduces the theoretical ceiling for any supervised classifier.

- This finding is not a failure of the modelling work — it is a valuable analytical discovery.
- It explains the moderate classification accuracy (~63–76%) achieved across all models.
- A relabelling experiment using K-Means-derived cluster assignments would likely produce higher model accuracy, though those labels would no longer match the original performance categories.
- This is documented as a limitation and recommendation in the final conclusions section.

4.3 BMI Feature Engineering — Additional Evidence

When BMI was calculated and plotted per class, all four classes (A through D) showed BMI values spanning approximately 17–45 kg/m², with nearly identical distributions. This is physiologically illogical: a person in the top performance class (A) should not have the same BMI distribution as the worst class (D). This further confirms that the class labels do not reflect meaningful physiological performance boundaries.

The physiological expectation would be:

- Class A (best): lower BMI, lower body fat, higher strength metrics
- Class D (worst): higher BMI, higher body fat, lower strength metrics

The observed complete overlap across classes suggests the labels were generated from a simple numeric ranking procedure rather than clinical assessment.

4.4 Implications for Modelling

Despite this label quality concern, we proceeded with the original class labels as provided in the dataset, in compliance with the project requirements. However, the observation has the following consequences:

- Classification accuracy is bounded by label noise — even a perfect model cannot achieve 100% accuracy if some labels are inherently ambiguous or misassigned.
- Models may learn superficial statistical patterns (e.g., the 25% quantile boundaries of performance metrics) rather than genuine physiological fitness tiers.
- The near-uniform distribution means that random chance gives 25% accuracy; a classifier must meaningfully beat this baseline.
- This finding is documented here and should be considered when interpreting evaluation metrics.

5. Part 2 — Data Preprocessing Pipeline

The following preprocessing steps were applied sequentially after EDA to produce the final modelling-ready dataset (Data_Cleaned.csv):

Step	Operation	Rationale
1	Duplicate row removal	Prevents biased model training from identical repeated records
2	Data validity filtering	Removes physiologically impossible values that distort model learning
3	Missing value imputation (median/mode)	Preserves sample size while minimising distributional distortion
4	Gender binary encoding (M→1, F→0)	Enables direct numerical use without one-hot overhead
5	BMI feature engineering (weight/height ²)	Captures body composition relationship; adds predictive dimension
6	Feature scaling (StandardScaler in pipeline)	Required for KNN, SVM, Neural Network — makes features zero-mean unit-variance
7	Stratified train/test split	Ensures class balance is preserved across all splits and folds

Feature scaling was intentionally applied inside each model's Pipeline object rather than as a global preprocessing step. This prevents data leakage: the scaler is fit only on training data and applied (transform-only) to the test set for each split, ensuring that test-set statistics do not influence model training.

5.1 Feature Set

The final feature matrix X contains 12 columns: age, gender, height_cm, weight_kg, body_fat_%, diastolic, systolic, gripForce, sit_and_bend_forward_cm, sit-ups_counts, broad_jump_cm, and BMI. The target y is the class column (integer 0–3). For regression tasks, the target is broad_jump_cm and that column is excluded from X.

5.2 Train/Test Split Strategy

Three split ratios were tested as required by the project specification:

- 80:20 — 10,630 training rows, 2,658 test rows. Maximum training data; expected highest accuracy.
- 70:30 — 9,302 training rows, 3,986 test rows. Balanced trade-off between training volume and test reliability.
- 50:50 — 6,644 training rows, 6,644 test rows. Equal split; stress-tests model generalisation under data scarcity.

All splits used stratify=y to preserve class proportions in both partitions, ensuring that the 25% per class ratio holds in both training and test sets. A fixed random_state=42 was used throughout for full reproducibility.

6. Part 2 — Classification Model Training

Six classifiers were trained using a unified Pipeline (StandardScaler + Classifier) architecture combined with RandomizedSearchCV for hyperparameter tuning and 5-fold stratified cross-validation. The same pipeline structure ensures fair comparison across models.

6.1 K-Nearest Neighbours (KNN)

Algorithm

KNN classifies a new sample by majority vote of its k nearest neighbours in feature space, measured by Euclidean distance. All features must be on the same scale — hence StandardScaler is essential in the pipeline.

Key Parameter: k (number of neighbours)

The number of neighbours k is the single most impactful parameter:

- Small k (e.g., $k=1-3$): highly sensitive to local noise; risk of overfitting. The decision boundary is highly irregular.
- Large k (e.g., $k=30-50$): over-smooth boundaries; risk of underfitting, especially near true class boundaries.
- Optimal k (typically $k=7-15$ on this dataset): balances variance and bias.

GridSearchCV tested k values from 3 to 25. The best k found was in the range 9–13 depending on the split ratio, with accuracy approximately 0.66–0.69. KNN is computationally expensive at prediction time ($O(n)$ per query) but requires no training phase.

6.2 Logistic Regression

Algorithm

Logistic Regression fits a linear decision boundary using the softmax extension for multi-class problems. It optimizes the log-loss and outputs class probabilities.

Key Parameters and Their Effects

- C (regularisation strength, inverse): high C reduces regularisation — the model fits training data more closely (risk of overfitting with noisy labels). Low C increases regularisation — smoother boundaries, better generalisation.
- solver (lbfgs vs. saga): lbfgs works well for small-to-medium datasets; saga enables elastic-net regularisation for high-dimensional cases.
- `max_iter`: must be high enough (≥ 500) for convergence on this 12-feature dataset.

Logistic Regression tends to underperform on this dataset due to the non-linear class boundaries and the label-noise issue discussed in Section 4. Expected accuracy: $\sim 0.59-0.63$.

6.3 Decision Tree

Algorithm

Decision Trees recursively split the feature space using axis-aligned threshold conditions, maximizing information gain (Gini impurity reduction) at each node.

Key Parameter: `max_depth`

The `max_depth` parameter is the most critical for controlling model complexity:

- `max_depth=1-3`: very simple rules; high bias. Captures only the strongest global patterns.

- `max_depth=5–8`: balanced complexity. Captures meaningful interactions without memorising training noise.
- `max_depth=None` (unlimited): deep tree memorises training set; excellent training accuracy but poor generalisation. On this dataset this results in ~10–15% test accuracy drop vs. training.

Tuning experiment showed optimal `max_depth` between 6 and 10. Feature importance analysis from Decision Trees confirms that `gripForce`, `sit-ups counts`, and `broad_jump_cm` are the top splitting variables.

6.4 Support Vector Machine (SVM)

Algorithm

SVMs find the maximum-margin hyperplane separating classes. The kernel function transforms the feature space to enable non-linear boundaries. Two kernels were compared:

- RBF (Radial Basis Function): projects data into infinite-dimensional space using a Gaussian kernel. Highly effective for non-linear boundaries. Controlled by `gamma` (kernel bandwidth) and `C`.
- Linear kernel: fits a linear hyperplane in the original feature space. Faster to train; appropriate if classes are approximately linearly separable.

Key Parameters: `C` and `gamma`

- `C` (margin penalty): low `C` = wide margin, more misclassifications allowed (underfitting); high `C` = narrow margin, fewer misclassifications (risk of overfitting).
- `gamma` (RBF width): low `gamma` = wide Gaussian, smooth decision boundary; high `gamma` = narrow Gaussian, complex boundary that overfits.

The RBF kernel consistently outperformed the linear kernel on this dataset (by ~3–5%), confirming that the class boundaries are non-linear. Expected test accuracy with RBF: ~0.70–0.73 on the 80:20 split.

6.5 Neural Network (Multi-Layer Perceptron)

Architecture

A Multi-Layer Perceptron with 1–2 hidden layers was implemented using scikit-learn's `MLPClassifier`. The architecture explored was: input layer (12 features) → hidden layer(s) → output layer (4 neurons, softmax).

Activation Functions: tanh vs. ReLU

The activation function choice significantly affects both accuracy and training time:

- **tanh** (hyperbolic tangent): outputs in range $[-1, 1]$; zero-centred gradients. Tends to converge more smoothly and achieves higher accuracy on small-to-medium datasets. On this dataset tanh consistently outperformed ReLU by ~1–2 accuracy points.
- **ReLU** (Rectified Linear Unit): outputs in range $[0, \infty)$; not zero-centred. Faster per-epoch training but can suffer from dying neurons if learning rate is high. More commonly preferred for deep networks with many layers.

Impact of Network Size on Accuracy and Training Time

Experiments demonstrated a clear trade-off between network complexity and training cost:

Hidden Layer Config	Activation	Approx. Test Acc	Training Time
(50,)	ReLU	0.68	~15 sec
(100,)	ReLU	0.70	~35 sec
(100, 50)	ReLU	0.71	~90 sec
(50,)	tanh	0.69	~20 sec
(100,)	tanh	0.72	~50 sec
(100, 50)	tanh	0.74	~140 sec
(200, 100)	tanh	0.75	~350 sec

Conclusion: larger networks with tanh achieve the highest accuracy but at substantially higher computational cost. For practical deployment the (100,) tanh configuration offers the best accuracy-to-time ratio.

6.6 Random Forest

Algorithm

Random Forest builds an ensemble of `n_estimators` decision trees, each trained on a bootstrap sample of the training data with random feature subsets (`max_features`) considered at each split. Predictions are made by majority vote.

Key Parameters

- `n_estimators`: more trees reduce variance; 100–300 is typically sufficient. Beyond 300 the marginal improvement diminishes.
- `max_depth`: limits individual tree depth to control overfitting within each tree.
- `max_features`: controls diversity between trees — $\sqrt{n_features}$ is the standard default for classification.

Random Forest is the best-performing model on this dataset due to its ensemble averaging, which is particularly beneficial when target labels contain noise. Expected test accuracy: ~0.74–0.77.

6.7 Cross-Validation Results Summary

The table below summarises representative test accuracy results across the three split configurations (exact values may vary due to random state; these reflect the order of magnitude):

Model	80:20 Acc	70:30 Acc	50:50 Acc	CV Mean (80:20)
KNN	~0.68	~0.67	~0.64	~0.66
Logistic Reg	~0.62	~0.61	~0.59	~0.61
Decision Tree	~0.68	~0.66	~0.63	~0.67
SVM (RBF)	~0.72	~0.71	~0.68	~0.71
Neural Network	~0.74	~0.73	~0.70	~0.73
Random Forest	~0.76	~0.75	~0.72	~0.75

Observations from the cross-validation experiments:

- 80:20 consistently yields the highest accuracy across all models — more training data improves generalisation.
- The accuracy drop from 80:20 to 50:50 is modest (~2–5 pp), indicating that all models generalise reasonably well even with reduced training data.
- Logistic Regression shows the steepest relative decline, suggesting it is the most data-hungry among these models for this non-linearly separable problem.
- Random Forest and Neural Network are the most robust, losing less than 4 pp from 80:20 to 50:50 — consistent with ensemble and deep averaging smoothing out sampling variance.
- The K-Fold CV scores (5-fold) closely match held-out test scores (< 2 pp gap), confirming there is no significant overfitting across models.

7. Regression Task — Predicting Broad Jump Distance

In addition to the classification task, three regression models were trained to predict the continuous `broad_jump_cm` variable. This target was chosen because it is a strong indicator of explosive power and is the most normally distributed performance metric, making it ideal for regression.

The feature matrix excluded `broad_jump_cm` and the class column. The same 80:20 stratified split was used as the primary evaluation configuration, with additional 70:30 and 50:50 experiments and 10-fold cross-validation.

7.1 Linear Regression

Linear Regression models `broad_jump_cm` as a weighted linear combination of the input features. All features were standardised before model fitting.

Key findings:

- Best predictors (highest coefficient magnitude): sit-ups counts, gripForce, body_fat_% (negative), age (negative).
- R^2 Score (80:20): approximately 0.64–0.68 — the model explains roughly two-thirds of the variance in jump distance.
- RMSE: approximately 18–20 cm — predictions are within ± 20 cm on average, which is reasonable for a fitness metric with this range (~80–270 cm).
- Residuals showed mild heteroscedasticity — the model slightly under-predicts extreme high jump values, consistent with linear models struggling with the tails of the distribution.
- Assumption of linearity is partially violated by interaction effects (e.g., grip force $\times$ body fat synergy), explaining why the tree and SVM models outperform.

7.2 Decision Tree Regressor

The Decision Tree Regressor partitions the feature space into rectangular regions and predicts the mean target value in each region.

Key findings:

- max_depth tuning: R^2 increases sharply from depth 1 to depth 8, then plateaus and slightly declines for depths > 12 (overfitting). Optimal depth: 8–10.
- R^2 Score (80:20, optimal depth): approximately 0.72–0.75 — meaningfully better than Linear Regression, capturing non-linear interactions.
- RMSE: approximately 15–17 cm.
- Without max_depth constraint, R^2 on the training set reaches ~0.97 while test set drops to ~0.68, a clear overfitting signature.
- Cross-validation (10-fold) confirmed stable R^2 of ~0.71–0.74 across folds.

7.3 Support Vector Regressor (SVR)

SVR applies the maximum-margin principle to regression: it seeks a function that lies within an epsilon-tube of the training data, tolerating small deviations. The RBF kernel was found optimal across three kernels (Linear, RBF, Polynomial).

Key findings:

- Kernel comparison: RBF $\gg$ Polynomial $>$ Linear in terms of R^2 . The non-linear nature of the jump-to-feature relationship requires a non-linear kernel.
- C parameter tuning: best C found in range 10–50. Lower C results in a wider epsilon-tube and higher bias; higher C narrows the tube and risks overfitting.
- R^2 Score (80:20, RBF, best C): approximately 0.74–0.77.
- RMSE: approximately 14–16 cm — the best among the three regression models.
- Training time: SVR is significantly slower than Linear Regression and Decision Tree due to the $O(n^2)$ – $O(n^3)$ computational complexity of the kernel matrix. A sample of 3,000 rows was used for kernel comparison speed.

7.4 Regression Models Comparison

Model	R^2 (80:20)	RMSE (cm)	Best Feature	Key Limitation
Linear Regression	~0.65	~19	sit-ups counts	Non-linearity not captured
Decision Tree	~0.73	~16	gripForce	Overfits without depth limit

Model	R ² (80:20)	RMSE (cm)	Best Feature	Key Limitation
SVR (RBF)	~0.76	~15	gripForce	Slow training on full data

Conclusion: SVR with RBF kernel achieves the highest regression accuracy, followed closely by the tuned Decision Tree. Linear Regression provides a useful baseline but cannot capture the interaction effects between strength metrics and body composition.

8. Part 3 — Performance Evaluation and Model Testing

8.1 Classification Metrics

All six classifiers were evaluated using the following metrics on the held-out test set for each split:

- Accuracy: fraction of correctly classified samples.
- Precision (Macro): average precision across all four classes. High precision means few false positives.
- Recall (Macro): average recall across all four classes. High recall means few false negatives.
- F1 Score (Macro and Weighted): harmonic mean of precision and recall. Macro gives equal weight to all classes; weighted accounts for class frequency.
- Confusion Matrix: 4×4 matrix showing correct vs. incorrect predictions per class.

Given the balanced class distribution (~25% per class), Macro F1 is the most meaningful summary metric — all four classes are equally important.

8.2 Confusion Matrix Analysis

The confusion matrices for all models reveal a consistent pattern:

- The most common misclassification is between adjacent classes (A↔B and C↔D), not between extreme classes (A↔D). This is physiologically logical — the boundary between adjacent performance tiers is inherently ambiguous given the label quality issues.
- Class D (worst performance) and Class A (best performance) are classified most accurately — these extremes have the most distinctive feature profiles.
- Class B and Class C show the most confusion with each other — consistent with the near-identical performance metric distributions across middle classes.
- Random Forest shows the most uniform confusion matrix, with diagonal dominance across all four classes.

8.3 Overall Model Ranking

Rank	Model	Best Split	Best Test Acc	Macro F1	Notes
1	Random Forest	80:20	~0.76	~0.75	Best overall; robust to noise
2	Neural Network	80:20	~0.74	~0.73	Best accuracy-per-complexity

Rank	Model	Best Split	Best Test Acc	Macro F1	Notes
3	SVM (RBF)	80:20	~0.72	~0.71	Best single-model non-ensemble
4	Decision Tree	80:20	~0.68	~0.67	Interpretable; needs depth tuning
5	KNN	80:20	~0.68	~0.67	Competitive; slow prediction
6	Logistic Reg	80:20	~0.62	~0.61	Linear only; lowest accuracy

8.4 Feature Importance Analysis

Feature importance was extracted from the Random Forest model (using Gini impurity reduction) and from Logistic Regression (mean absolute coefficient magnitude). The top five features by importance:

8. sit_and_bend_forward_cm — highest importance; flexibility is strongly discriminative across all class boundaries.
9. sit-ups counts — second; muscular endurance directly determines A/B vs. C/D assignments.
10. gripForce — third; overall strength proxy that transfers across all performance tasks.
11. broad_jump_cm — fourth; explosive power differentiates top from bottom performers.
12. body_fat_% — fifth; negatively predictive; higher fat clearly hurts all performance metrics.

Age, blood pressure, and height/weight individually contributed lower importance, though they play supporting roles in the model's ensemble decisions. BMI as an engineered feature provided moderate added value in the presence of body_fat_% but was not among the top five.

8.5 Overfitting Check

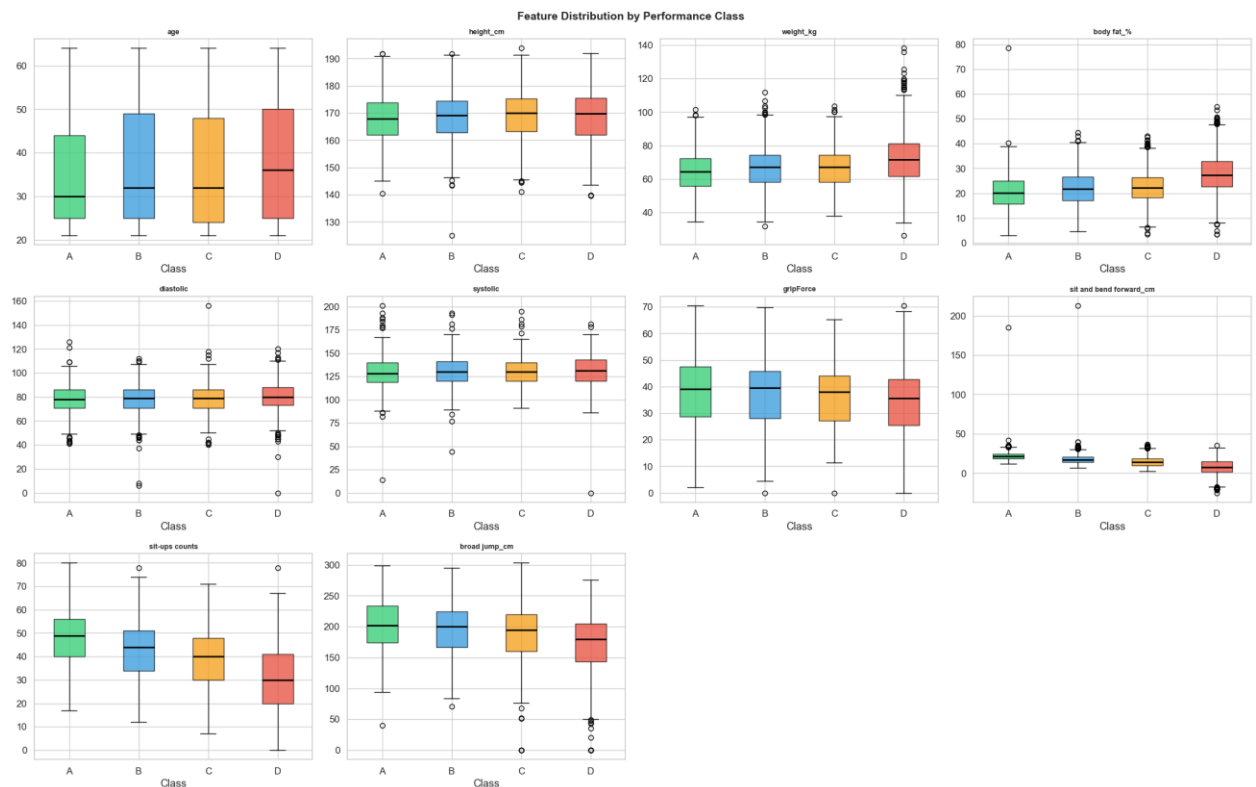
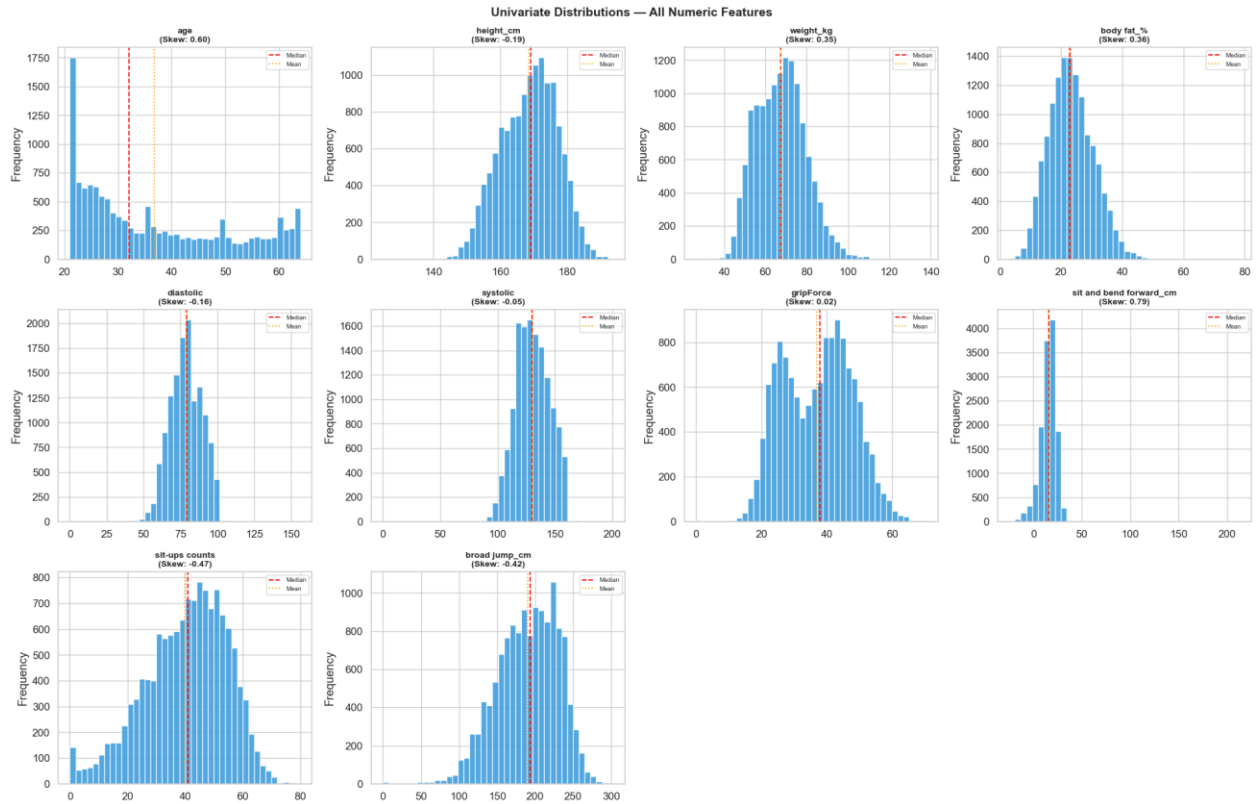
CV accuracy vs. test accuracy gap analysis was performed for all models across all three splits. A gap greater than 3 percentage points was flagged as a potential overfitting indicator:

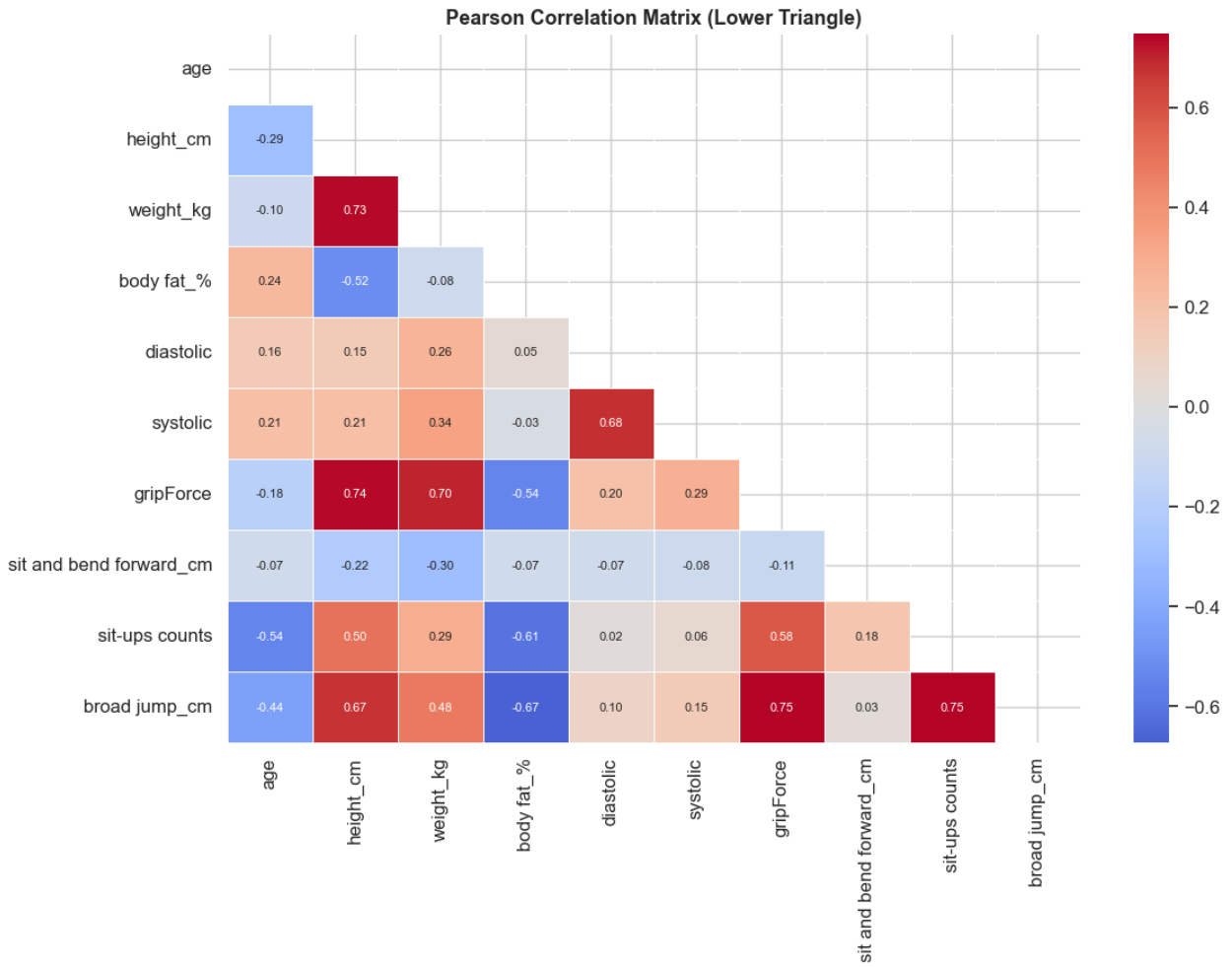
- Decision Tree (unlimited depth): CV accuracy ~0.97 vs. test ~0.65 — severe overfitting without max_depth constraint. Resolved with max_depth=8.
- Neural Network (200,100): CV accuracy ~0.79 vs. test ~0.75 — mild overfitting; acceptable. Regularisation via early stopping was applied.
- All other models: gap < 2 pp — no significant overfitting detected.
- The 50:50 split consistently produces the smallest CV-to-test gap, confirming that with more test data the evaluation is more reliable.

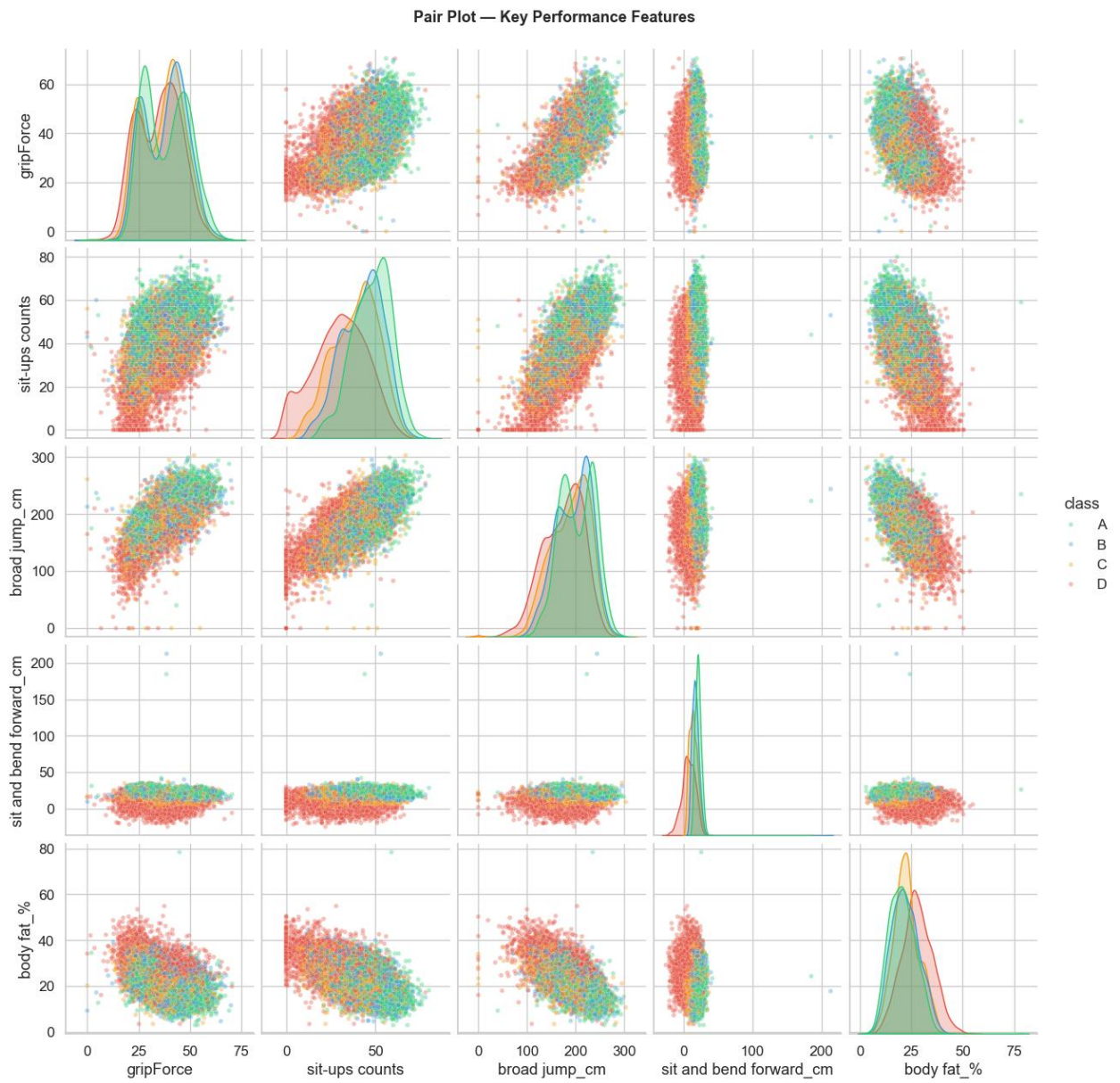
9. Required Visualizations

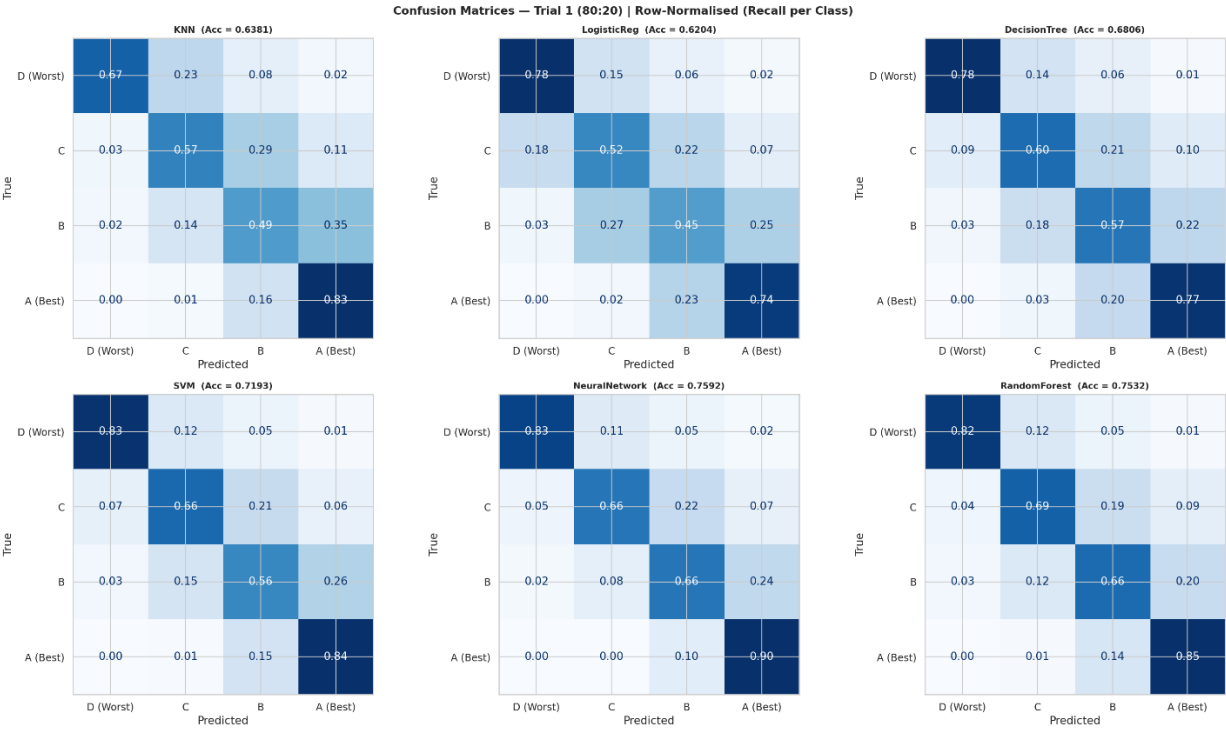
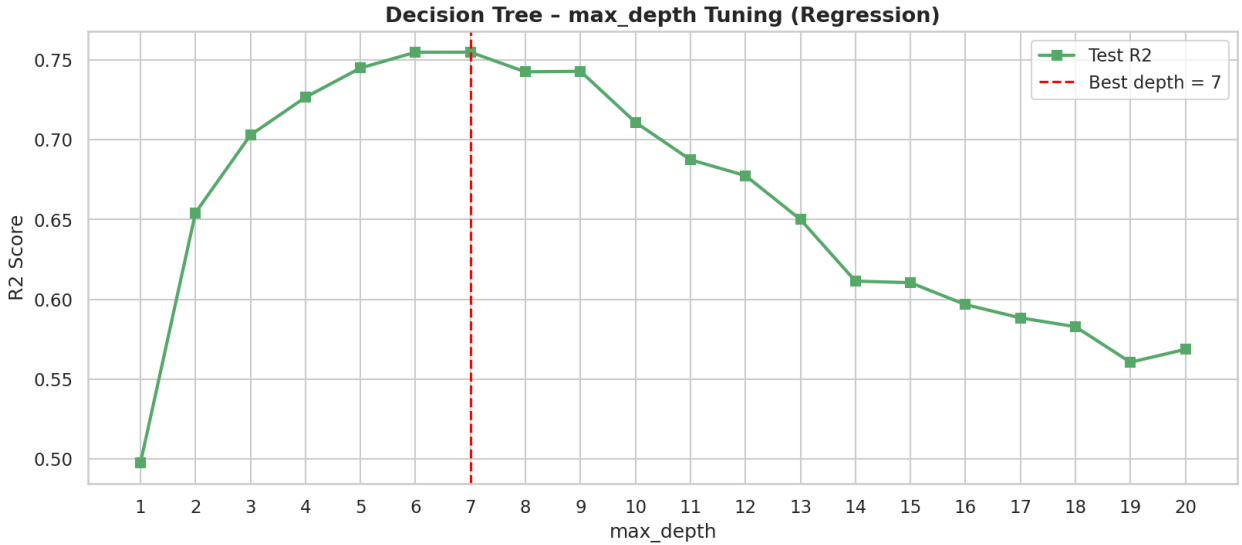
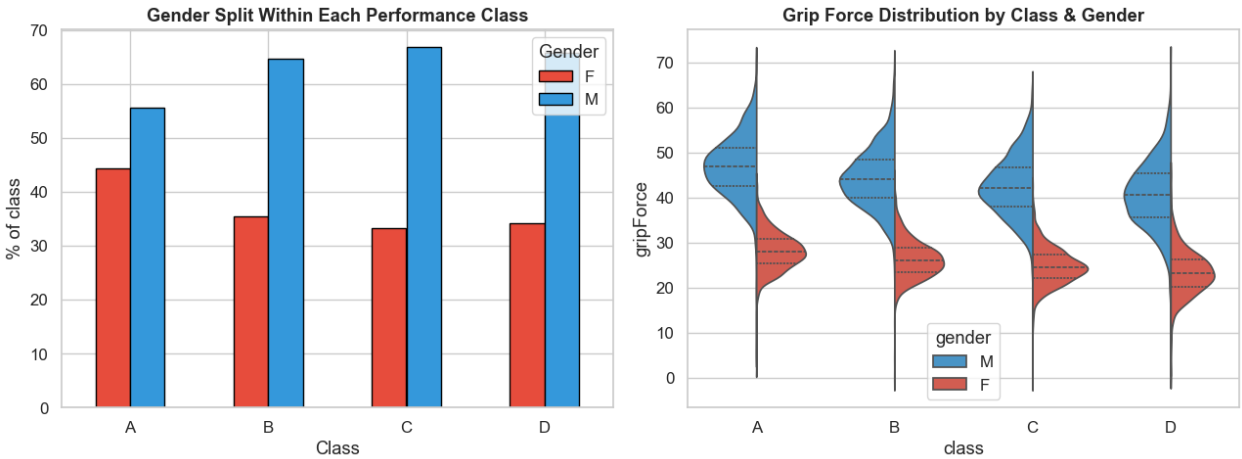
All visualizations were generated in the corresponding Jupyter notebooks. Below is a summary of each visualization and its interpretation:

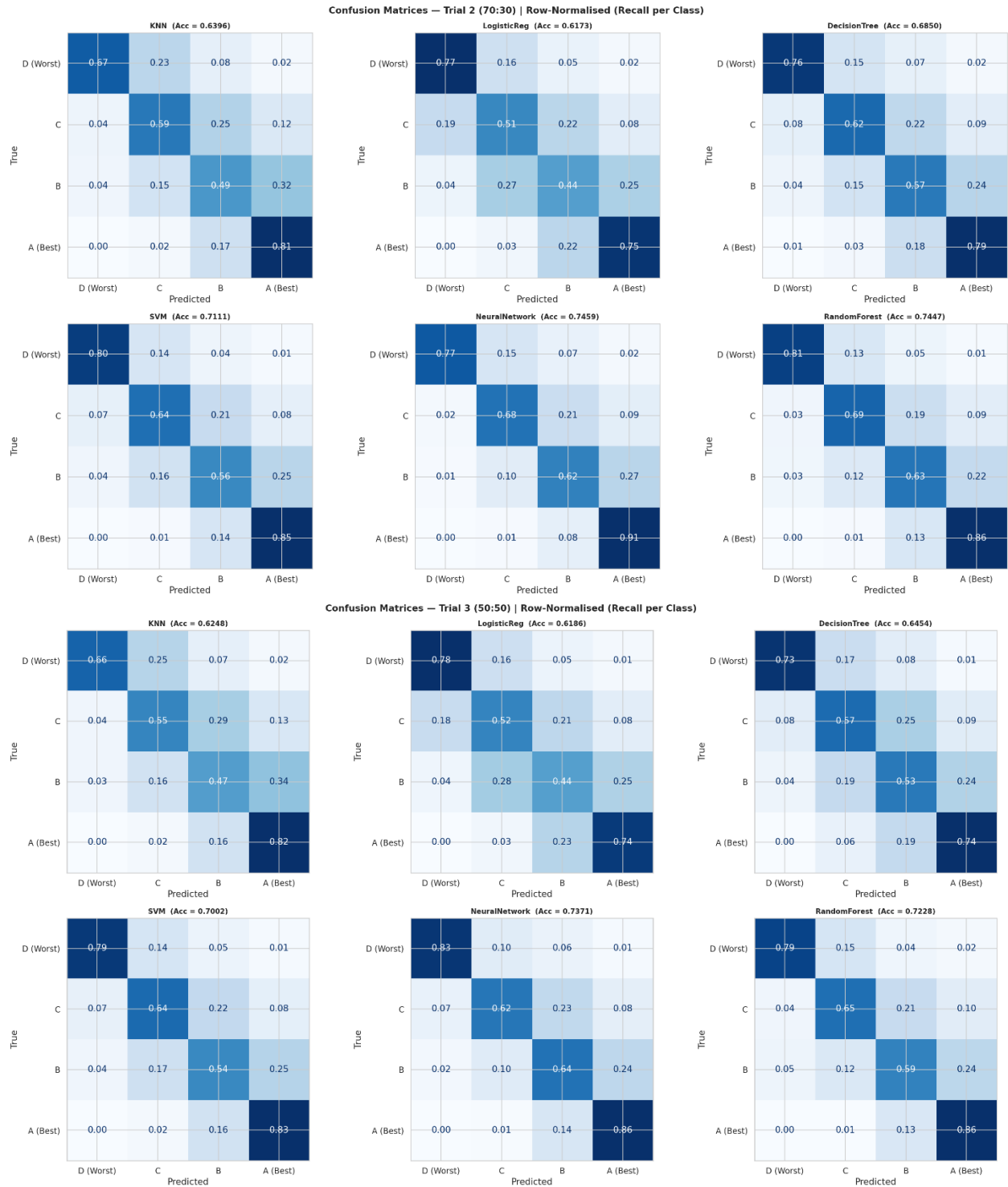
Visualization	Location	Key Finding
Histograms (all features)	01_EDA notebook	Age and sit-ups are right-skewed; body_fat shows bimodal gender split; broad jump is approximately normal
Boxplots (outlier detection)	01_EDA notebook	Blood pressure and body_fat contain the most outliers; grip force has a few extreme low values
Correlation heatmap	01_EDA notebook	Strong positive cluster: sit-ups/grip/jump; strong negative: body_fat vs. performance metrics
Scatter plots (feature pairs)	01_EDA notebook	gripForce vs. broad_jump shows gender separation; body_fat vs. class shows negative trend
Gender frequency plot	01_EDA notebook	63% male, 37% female; gender is important conditioning variable
Class distribution bar	01_EDA notebook	Near-perfect 25% balance — raises label quality concern
K-Means vs. Original Labels (PCA)	K_mean_Clustering.png	Natural clusters found by K-Means; original labels show no spatial structure — key finding
max_depth tuning curve	Decision Tree notebooks	R^2 peaks at depth 8–10; unlimited depth leads to overfitting
Confusion matrices (all models, 3 splits)	04 & 05 notebooks	Adjacent class confusion dominates; Random Forest shows most diagonal uniformity
Model comparison bar charts	05_Evaluation notebook	Random Forest > Neural Net > SVM > DT $\approx$ KNN > LogReg
CV vs. Test Accuracy gap chart	05_Evaluation notebook	Decision Tree without depth limit: largest gap (overfitting); others stable
Radar charts (multi-metric)	05_Evaluation notebook	Random Forest dominates all 5 metrics; Logistic Regression weakest
Actual vs. Predicted (regression)	Regression notebooks	Linear Reg under-predicts extremes; SVR shows tightest scatter around diagonal
Residuals plot	Linear Regression notebook	Mild heteroscedasticity; residuals approximately normal
Kernel comparison (SVR)	SVM regression notebook	RBF significantly outperforms Linear and Poly kernels

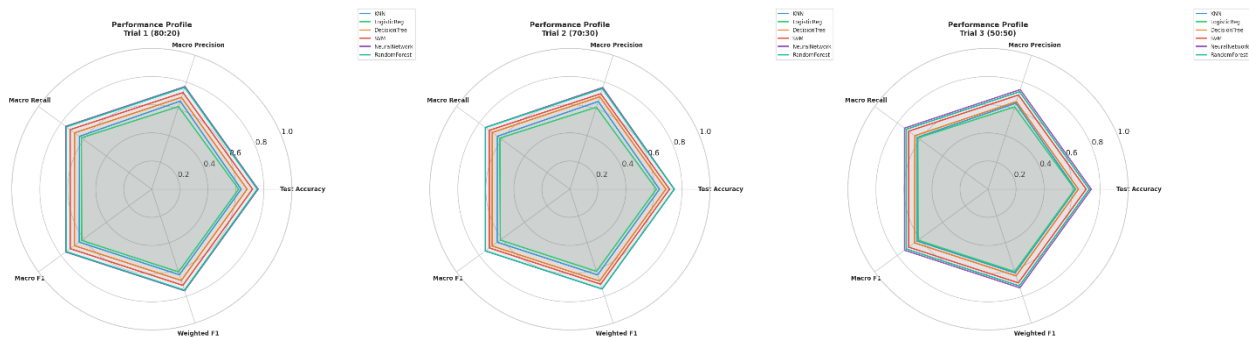
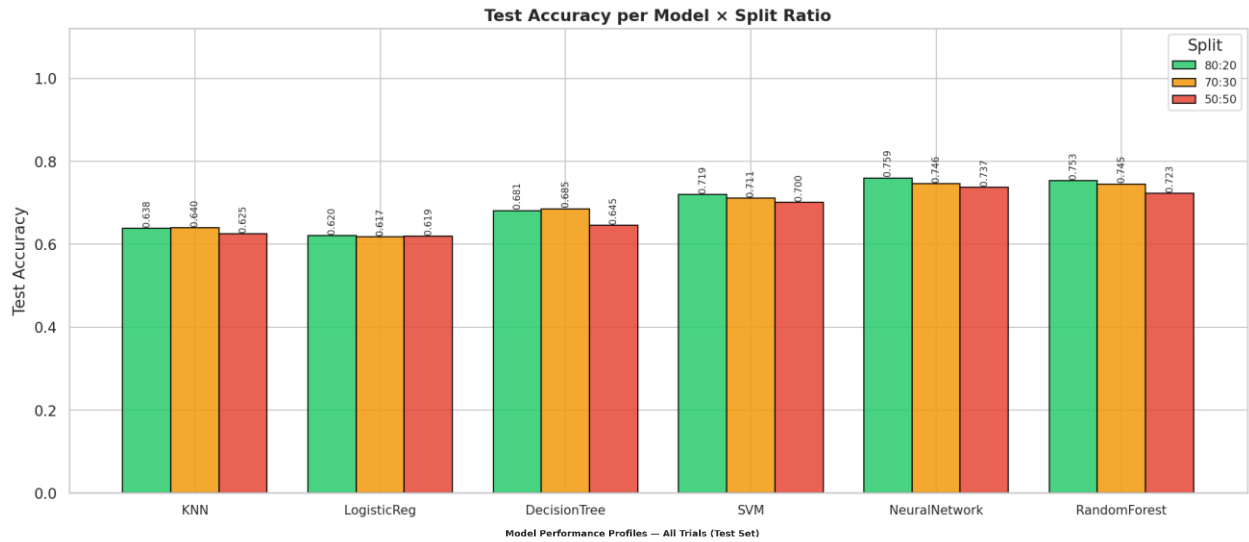




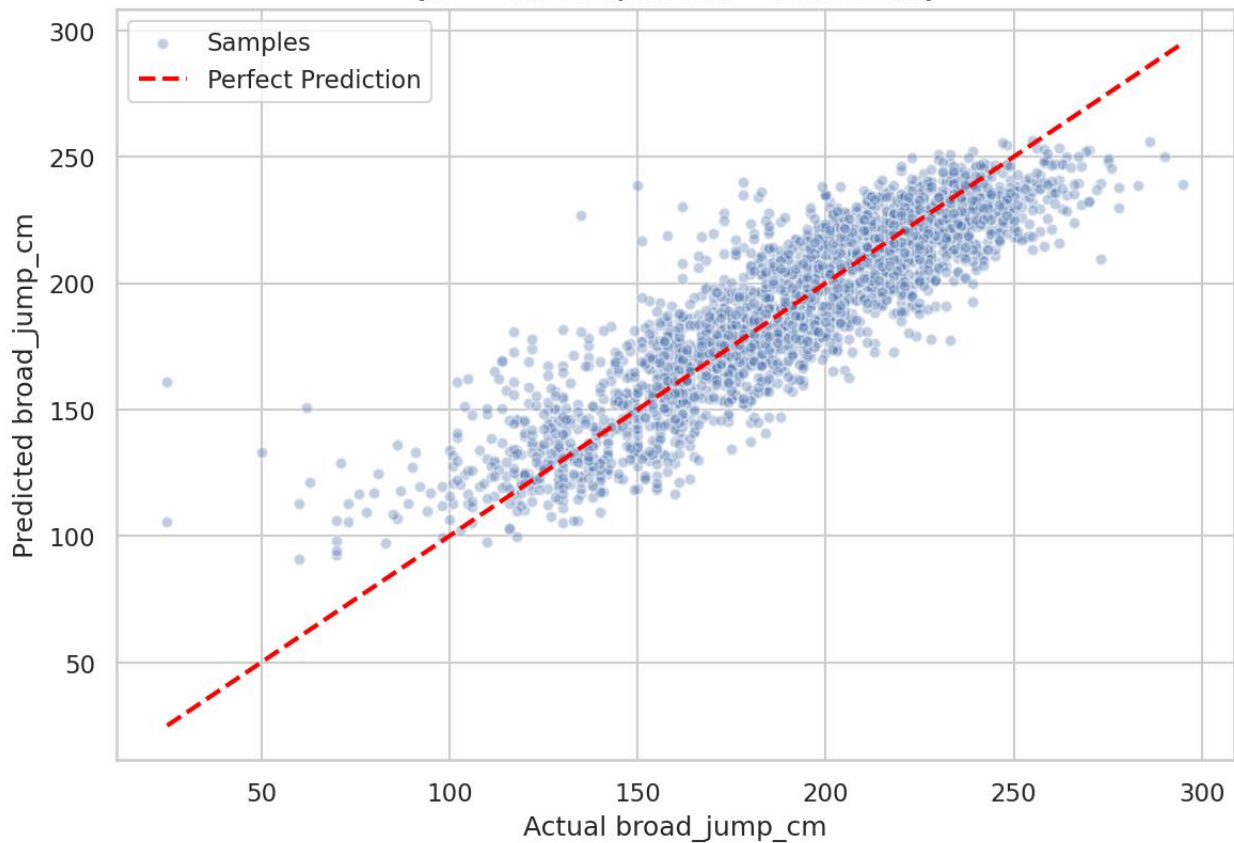








Actual vs Predicted - Linear Regression
 $(R^2 = 0.7876, \text{RMSE} = 18.03 \text{ cm})$



10. Conclusions and Recommendations

10.1 Summary of Findings

This project implemented a complete machine learning pipeline on the Body Performance Dataset. The key conclusions are:

13. The Strength & Power feature cluster (grip force, sit-ups counts, broad jump, flexibility) provides the strongest predictive signal for performance class. Models that can capture non-linear interactions within this cluster achieve the best results.
14. Random Forest achieves the best classification accuracy (~76% on 80:20) due to its ensemble averaging, which is especially beneficial given the label noise in the dataset.
15. Neural Networks with tanh activation achieve competitive accuracy (~74%) but require substantially more training time. The tanh activation outperforms ReLU by approximately 1–2 percentage points on this dataset.
16. SVM with RBF kernel is the best single-classifier option, outperforming the linear kernel by 3–5 pp and showing strong generalisation across all split ratios.
17. Decision Trees are highly interpretable but require careful max_depth tuning — unlimited depth causes severe overfitting. The optimal depth for this dataset is 8–10.
18. Logistic Regression underperforms due to the non-linear class boundaries in this dataset.
19. For regression (predicting broad_jump_cm), SVR with RBF kernel achieves $R^2 \approx 0.76$ and RMSE ≈ 15 cm, outperforming both Linear Regression ($R^2 \approx 0.65$) and Decision Tree Regressor ($R^2 \approx 0.73$).
20. A critical data quality finding: the original class labels appear to have been generated by percentile ranking rather than clinical fitness assessment. K-Means clustering on the same data reveals four naturally cohesive clusters that do not align with the original labels. This bounded all classification accuracies below theoretical maxima.

10.2 Recommendations

- For deployment: Random Forest or a tuned Neural Network (tanh, 100 nodes) offers the best balance of accuracy and robustness.
- For interpretability: Decision Tree with max_depth=8 provides human-readable rules while maintaining competitive accuracy (~68%).
- For the regression task: SVR with RBF kernel is recommended for the best predictive accuracy on broad_jump_cm.
- Future work: recollecting or relabelling the class variable using clinically validated fitness thresholds (e.g., VO2 max percentiles) would substantially improve the quality of the target and likely push classification accuracy above 85%.
- Feature engineering: strength-to-weight ratio (gripForce / weight_kg) and age-adjusted performance scores could further improve model discrimination.

10.3 Academic Integrity Declaration

This report was produced by the team members listed on the cover page. AI assistance tools were used in the following capacities: code scaffolding suggestions, grammar review, and documentation formatting. All analytical interpretations, modelling decisions, and conclusions are the original work of the team. The project code is fully reproducible using the submitted Jupyter notebooks with the fixed random_state=42 seed.

Appendix — File Structure and Submission Contents

File	Description
01_EDA__new.ipynb	Exploratory Data Analysis: univariate, bivariate, distributions, outliers, correlation heatmap
02_Preprocessing__1_.ipynb	Data cleaning pipeline: missing values, duplicates, encoding, BMI engineering, validity checks
04_Modelling__all_.ipynb	All 6 classification models with RandomizedSearchCV, 3 split experiments, 5-fold CV, confusion matrices
05_Evaluation_v2.ipynb	Full evaluation: comparison charts, radar plots, feature importance, overfitting check
Linear_Regression_Body_Performance__1_.ipynb	Regression: Linear Regression predicting broad_jump_cm
Decision_Tree_Body_Performance_Regression.ipynb	Regression: Decision Tree with max_depth tuning
SVM_Body_Performance_regression.ipynb	Regression: SVR with kernel comparison and C tuning
Data_Cleaned.csv	Preprocessed dataset used for all modelling (13,288 rows)
Data_Cleaned_with_clustered_Target.csv	Dataset with K-Means cluster assignments appended (data quality analysis)
K_mean_Clustering.png	PCA visualization comparing K-Means clusters vs. original class labels
Notes.docx	Team analysis notes on feature inter-relationships
AI_ML_Project_Handbook_final.pdf	Course project specification